Course Title: Technology, Environment and Society

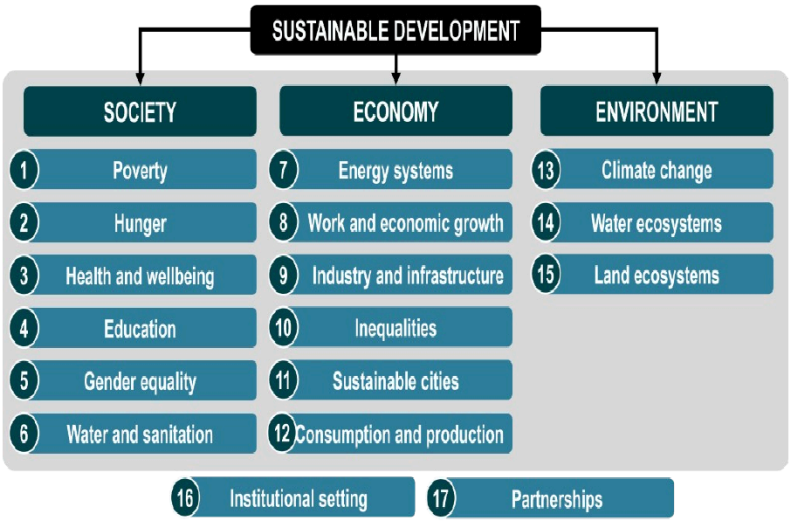
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Content

- Sustainability and Sustainable Development
- Environmental Stewardship
- Environmental Justice and Equity

Sustainability and Sustainable Development





- Each of these goals can be associated with societal, economic, or environmental improvements:
- **Society**. These goals prioritize satisfying the diversified needs of the population, such as food, health, and education being among the most basic. Maintaining human capital (knowledge, skills, and capabilities) is mostly the responsibility of educational systems, but corporations also provide substantial training opportunities to their workforce.
- **Economy**. These goals promote improvements in the welfare of populations. Key concepts are related to achieving or sustaining economic growth, maximizing profits, increasing competitiveness, and expanding markets. Globalization has given a new dimension to economic development by enabling an extended range of comparative advantages. However, like all economic processes, globalization promotes growth differently as regions and social classes capture their opportunities differently. This has led to inequalities.
- **Environment.** These goals concern the footprint of human activities on environmental systems, notably their carrying capacity. The overarching issue of climate change remains salient, particularly over the issue of carbon emissions.

Sustainability and **Sustainable Development**

5 Important Measures for Sustainable Development

(i) Technology

Using appropriate technology is one which is locally adaptable, eco-friendly, resource efficient and culturally suitable.



This concept is known as "design with nature"

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Sustainability and Sustainable Development

(ii) Reduce, Reuse, and Recycle Approach

The 3-R approach advocating minimization of resource use, using them again and again instead of passing it on to the waste stream

(iii) Promoting Environmental Education and Awareness

Making environmental education the centre of all learning process will greatly help in changing the attitude of people towards the earth and the environment.

feeling of belongingness to earth

'Earth thinking'

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Sustainability and Sustainable Development

(iv) Resource Utilization as Per Carrying Capacity:

If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts and continues till it reaches a point of no return.

Carrying capacity has two basic components:

i. Supporting capacity i.e. the capacity to regenerate
ii. Assimilative capacity i.e. the capacity to tolerate different stresses.

(v) Improving Quality of Life Including Social, Cultural and Economic Dimensions:

Ensuring the distribution of fare share of benefits

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Environmental Stewardship

Individual or Organizations practice:-

✓ Use resources efficiently
✓ Reduce waste and minimize the negative impacts on the environment
✓ Translate reduced consumption and waste into financial savings
✓ Provide a healthy, more efficient and effective working environment

Environmental Stewardship

✓ Embodies the concept of a Green Office, a Green School or Green Home which is conceptualized as smarter and better as it is:

❑ Ecological - using non-toxic, recycled, environmentally friendly products and supplies
❑ Efficient – using as little energy and other resources as possible and putting out the smallest amount of waste as possible
❑ Healthy – will generate as little visual, noise and physical pollution as possible

Environmental Stewardship

Principles of Environmental Stewardship

Principle 1: Resource Conservation – making the most efficient use of resources (materials, energy and water) through:

- Energy Efficiency and Conservation
- Water Efficiency and Conservation
- Solid Waste Management and Conceptualizing Waste as a Resource

Principle 2: Pollution Prevention – minimizing contamination of the environment by chemicals or other materials through:

- Vehicle/Fleet Management
- Management of the Use of Chemicals and Other Hazardous Substances

Principle 3: Occupational Health and Safety

- Managing Indoor Air Quality
- Comfort and Productivity in the Workplace
- Emergency and Disaster Management

Environmental Justice and Equity

Fair treatment means

….that no group of people should bear a disproportionate share of the negative environmental consequences resulting from industrial, governmental and commercial operations or policies.

Environmental Justice and Equity

Meaningful involvement means

- people have an opportunity to participate in decisions about activities that may affect their environment and/or health;
- the public contribution can influence the regulatory agency's decision;
- their concerns will be considered in the decision making process;
- the decision makers seek out and facilitate the involvement of those potentially affected.

Environmental Justice and Equity

What is the Problem?

- The impact of environmental policies and decisions have differentiated impacts across population groups
- Most countries have utterly no idea what environmental justice is nor have any mechanism for addressing EJ issues

Examples

- Haiti and much of the developed Western Hemisphere
- Environmental neglect in New Orleans
- The former East German Republic and its neighbors
- Philippines Islands and Japan
- Sub-Saharan Africa and Europe